

What actually makes data agents cheaper?

Four workloads. Real Gemini calls. Quality measured before cost.

This report explains the recorded experiments in plain language. It compares execution policies on the same held-out tasks and separates useful savings from results that are not ready to adopt.

72% lower model cost for incremental refresh

Sending only changed fields cost about **\$0.00831 per 1,000 accepted results**, versus **\$0.02971** for one-item Flash Lite execution. Both passed 300/300 held-out records. These are usage-based API estimates; optimization overhead is separate.

The strongest simple baseline still wins

A deterministic parser passed all four workloads without a model call. CPU cost is not priced. These source records are structured enough to parse, so the experiments do not establish a proprietary advantage or superiority to Amazon AgentCore.

Workload	Frozen router accepted	Adoption result
Invoice & order extraction	270/300	Hold
Public project research	300/300	Hold
Research metadata collection	300/300	Hold
Incremental data refresh	300/300	Hold

All four frozen routers remain unapproved: they either fail quality/latency gates or do not improve cost over the parser. A held-out test is an audit, not a chance to reselect the winner.

A fair comparison starts with the task.

The experiment in six steps

1. Freeze 2,000 input cases: four workloads, each with 100 development, 100 validation and 300 held-out test cases.
2. Use development cases to inspect failure modes and adapt published Strands reflection methods to Gemini.
3. Compare policies on validation cases. Freeze routing by input segment, with baseline fallback where evidence is sparse.
4. Execute every policy on held-out cases. Evaluate exact reference fields; charge failed attempts and fallback calls too.
5. Apply adoption gates: at least 95% observed acceptance, no observed quality drop, p95 total latency no higher than 60 seconds, complete priced usage, and strictly lower cost per accepted result.
6. Report 36 additional traffic stress runs across single/concurrent and steady/burst arrivals. No browser click launches a paid API call.

What the table columns mean

Accepted: outputs matching the required reference fields. **USD / 1k accepted:** total estimated model-call cost, including failures, divided by accepted outputs and multiplied by 1,000. **p95 total:** 95% of items finished within this time, including queueing. The main test submits 300 items in a burst.

What is included - and what is not

Prices use standard Gemini text rates, with known cached-read tokens priced separately. They are estimates, not reconciled account charges. Common development/validation acquisition and method-specific reflection costs are reported separately. Missing billing information stays unresolved. CPU, human labeling and initial acquisition of previous update state are not monetized.

Independent workloads overlapped on the same provider account. Latency is observational, not proof of a causal speedup. Public metadata was reformatted and stress conditions constructed; this is not customer production traffic.

Invoice & order extraction

Extract five invoice fields while ignoring stale drafts and injected footer instructions. Invoices are constructed test records.

Execution policy	Accepted	USD / 1k accepted	p95 total
Flash, one item per call	126/300	\$0.93401	172.17s
Flash Lite, one item per call	295/300	\$0.06432	98.67s
Flash Lite, batches of 4	271/300	\$0.07424	44.33s
Flash Lite, batches of 16	300/300	\$0.05934	23.23s
Rule-based model escalation	264/300	\$0.16279	46.90s
Deterministic source parser	300/300	\$0.00000	0.32s
Reins frozen routing	270/300	\$0.18987	59.73s

What this means

The frozen router accepted 270 of 300 records. Amount strings sometimes became numbers, which fails the required output format. It should not replace the comparator. Batches of 16 passed all cases, but the parser also passed without model calls.

Adoption decision: hold. quality below the required acceptance bar; no cost improvement over the parser.

Optimization and comparison context

Shared development/validation acquisition: approximately **\$0.08149**. This is additional to the per-result execution costs above.

Single reflector: failed to produce an accepted prompt; 31 model calls. Multi reflector: failed to produce an accepted prompt; 6 model calls.

Strands methods were adapted to the same Gemini provider; no managed AWS service was benchmarked. Failed adaptations are not evidence of inferior AWS performance. The online case page contains traffic results, example outputs and overhead amortization.

[Open the interactive case report](#) | [AgentCore documentation](#) | [Gemini pricing](#)

Public project research

Collect repository language, license, default branch and archive status from reformatted GitHub metadata.

Execution policy	Accepted	USD / 1k accepted	p95 total
Flash, one item per call	299/300	\$0.29247	163.39s
Flash Lite, one item per call	300/300	\$0.05513	120.92s
Flash Lite, batches of 4	232/300	\$0.09030	96.51s
Flash Lite, batches of 16	275/300	\$0.06198	41.37s
Rule-based model escalation	300/300	\$0.14737	57.86s
Deterministic source parser	300/300	\$0.00000	0.33s
Strands: single-reflector prompt	294/300	\$0.05815	44.66s
Strands: multi-reflector prompt	231/300	\$0.08854	104.72s
Reins frozen routing	300/300	\$0.14874	78.94s

What this means

The frozen router passed all 300 cases, but missed the 60-second p95 gate and cost more than the parser. Some batched alternatives lost accuracy; larger batches are not automatically a better execution policy.

Adoption decision: hold. p95 latency above 60 seconds; no cost improvement over the parser.

Optimization and comparison context

Shared development/validation acquisition: approximately **\$0.09615**. This is additional to the per-result execution costs above.

Single reflector: completed; 33 model calls. Multi reflector: completed; 19 model calls.

Strands methods were adapted to the same Gemini provider; no managed AWS service was benchmarked. Failed adaptations are not evidence of inferior AWS performance. The online case page contains traffic results, example outputs and overhead amortization.

[Open the interactive case report](#) | [AgentCore documentation](#) | [Gemini pricing](#)

Research metadata collection

Collect title, author, publication year, DOI and publication type from reformatted Crossref records.

Execution policy	Accepted	USD / 1k accepted	p95 total
Flash, one item per call	300/300	\$0.50033	239.91s
Flash Lite, one item per call	296/300	\$0.07225	75.60s
Flash Lite, batches of 4	244/300	\$0.10409	55.22s
Flash Lite, batches of 16	298/300	\$0.06557	34.31s
Rule-based model escalation	300/300	\$0.18271	79.04s
Deterministic source parser	300/300	\$0.00000	0.33s
Strands: single-reflector prompt	174/300	\$0.17669	101.64s
Strands: multi-reflector prompt	249/300	\$0.13010	72.05s
Reins frozen routing	300/300	\$0.22041	61.06s

What this means

The frozen router passed all 300 cases, but missed the latency gate and did not beat the parser on model cost. These structured records do not yet test the harder task of reading arbitrary research websites.

Adoption decision: hold. p95 latency above 60 seconds; no cost improvement over the parser.

Optimization and comparison context

Shared development/validation acquisition: approximately **\$0.13314**. This is additional to the per-result execution costs above.

Single reflector: completed; 22 model calls. Multi reflector: completed; 13 model calls.

Strands methods were adapted to the same Gemini provider; no managed AWS service was benchmarked. Failed adaptations are not evidence of inferior AWS performance. The online case page contains traffic results, example outputs and overhead amortization.

[Open the interactive case report](#) | [AgentCore documentation](#) | [Gemini pricing](#)

Incremental data refresh

Refresh controlled changes to project records. Every policy receives the same previously verified state.

Execution policy	Accepted	USD / 1k accepted	p95 total
Flash, one item per call	300/300	\$0.18806	107.74s
Flash Lite, one item per call	300/300	\$0.02971	80.95s
Flash Lite, batches of 4	300/300	\$0.02687	37.48s
Flash Lite, batches of 16	300/300	\$0.02743	17.04s
Rule-based model escalation	300/300	\$0.02687	32.13s
Changed fields only	300/300	\$0.00831	25.11s
Deterministic source parser	300/300	\$0.00000	0.31s
Strands: single-reflector prompt	300/300	\$0.02565	34.09s
Strands: multi-reflector prompt	300/300	\$0.02547	31.71s
Reins frozen routing	300/300	\$0.00000	0.32s

What this means

Changed-fields-only execution passed 300 of 300 cases and cut estimated model cost by 72% versus one-item Flash Lite calls. The parser also passed with no model calls. The gain supports incremental execution, not a claim of exclusive technology.

Adoption decision: hold. no cost improvement over the parser.

Optimization and comparison context

Shared development/validation acquisition: approximately **\$0.06162**. This is additional to the per-result execution costs above.

Single reflector: completed; 9 model calls. Multi reflector: completed; 26 model calls.

Strands methods were adapted to the same Gemini provider; no managed AWS service was benchmarked. Failed adaptations are not evidence of inferior AWS performance. The online case page contains traffic results, example outputs and overhead amortization.

[Open the interactive case report](#) | [AgentCore documentation](#) | [Gemini pricing](#)